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OPTIMUS PRIME / G1 EXPEDITION

A SLIP BECOMES A RESCUE.

G1 MUST RECOVER BEFORE A HUMAN MUST INTERVENE.

Extreme terrain turns small failures into high-consequence missions.



FOUR SKILLS FOR EXTREME TERRAIN.

ONE EXPEDITION SYSTEM



01 SELF-ARREST

STOP THE
SLIDE



02 FALL RECOVERY

GET BACK
UP



03 CONTROLLED RAPPEL

DESCEND WITH
CONTROL



04 TEAM LIFT

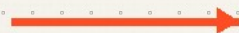
MOVE HEAVIER
LOADS TOGETHER

ONE POLICY. MANY FALLS.

SINGLE-ROBOT PPO

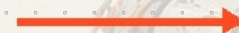
125 OBSERVATIONS

pose · speed · tool load



PPO 2×256

compact learned policy



14 ARM COMMANDS

plant and load the axe

EASY SLIDES → 5 m/s + CROSS-SLOPE FALLS

SUCCESS COUNTS ONLY WHEN THE AXE PHYSICALLY BITES.



ONE SHARED POLICY. EVERY ROBOT.

COOPERATIVE MAPPO



LOCAL STATE

+ TEAMMATE TOKENS



SHARED MAPPO

10 COMMANDS / G1

Same actor weights on every robot.

LIFT FIRST → CARRY + TURN → HEAVIER LOADS

LIVEKIT IS THE EXPEDITION BUS.

STATE + VOICE

STATE

ONE STATE STREAM / G1



LIVEKIT ROOM



EVERY ACTOR STAYS IN SYNC

VOICE

"MOVE THE LOG"



CONFIRM



GATED START INTENT

VOICE SUPERVISES. LOCAL POLICIES CONTROL MOTION.

FOUR SKILLS. MEASURED RESULTS.

FROZEN EVALUATION

ARRESTS PASSED

69 / 69

9 named + 60 unseen

ICY FIXED LINE

STANDS AGAIN

recover + continue

RAPPEL DESCENT

2.00 m

controlled braking

TARGET LOAD SPLIT

50% / 50%

balanced teamwork

STOP · RECOVER · DESCEND · COORDINATE